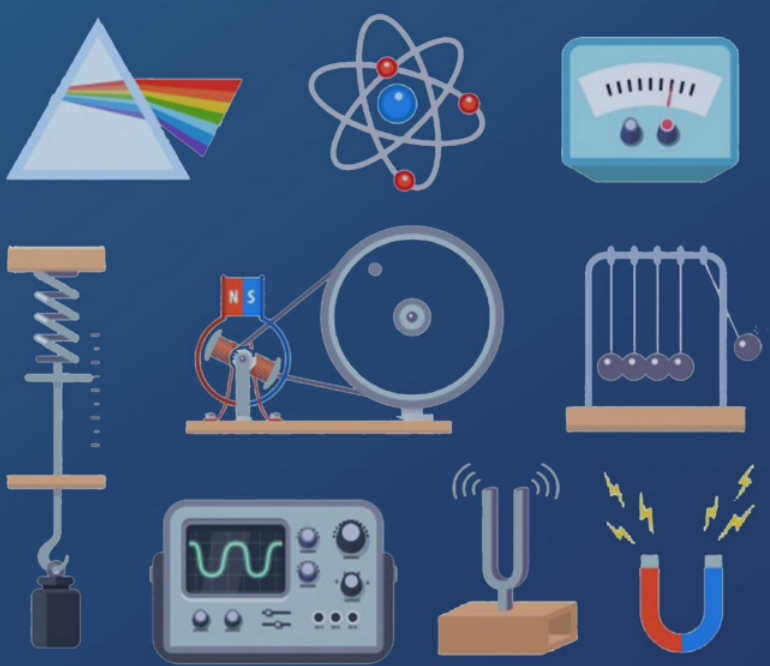
PHYSICS

Fluid Mechanics



SHEET

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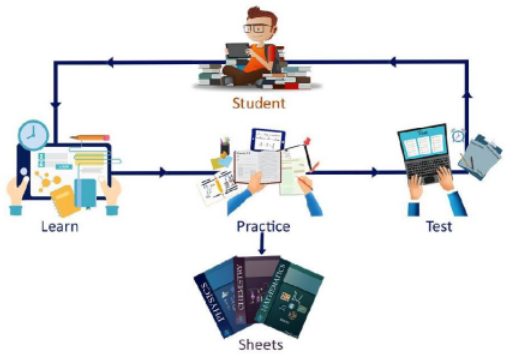
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Pressure				
2	Why is Liquid Surface Horizontal?				
3	Pressure Variation	1-3	1		
4	U-Tube and Barometer	4	11	1	
5	Pascal's Law and Hydrostatic Paradox		2, 12	2-4	1
6	Buoyant Force	5-6	3	5	2
7	Archimedes Principle	7-11	13	6-11	3
8	Eureka-I Found It	12	9	12-14	4
9	Fluid Dynamics	13-14			
10	Bernoulli's Principle	15		15	21
11	Application of Bernoulli's principle	16		16	
12	Speed of Efflux	17	4	17-22	5-9
13	Viscosity	18-19	5, 14	23	10
14	Stokes' Law	20		24	
15	Surface Tension	21-22		25-26	11, 22
16	Excess Pressure in Drops and Bubbles	23	6-7, 10	27	12
17	Contact Angle	24			13
18	Rise of Liquid in Capillary	25		28	14, 23
19	After full chapter		8	29	15-20

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Pressure Variation

Q1. A closed compartment containing gas is moving with some acceleration in horizontal direction. Neglect effect of gravity. Then, the pressure in the compartment is :–

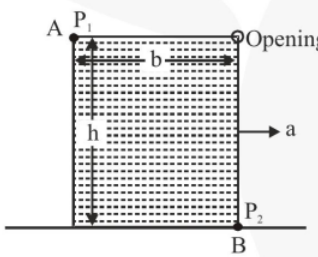
(A) same every where

(B) lower in front side

(C) lower in rear side

(D) lower in upper side

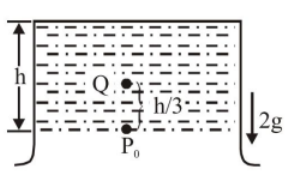
Q2. A closed rectangular vessel completely filled with a liquid of density ρ moves with an acceleration a = g. The value of the pressure difference at point A and B i.e. $(P_1 - P_2)$ is :



(A) ρgb (B) $\frac{\rho g(b+h)}{2}$

(C) ρ(ab – gh) (D) ρgh

Q3. An inverted test tube having a liquid of density ρ accelerates down with a = 2g then, $P_Q - P_0$ is:

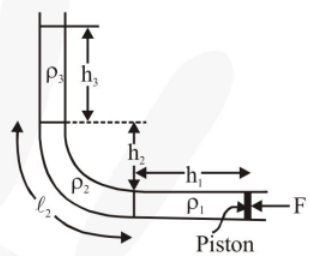


(A) $-\frac{\rho gh}{3}$ (B) $\frac{2\rho gh}{3}$

(C) $\frac{\rho gh}{3}$ (D) indeterminate

U-Tube and Barometer

Q4. A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in equilibrium by the piston of area A, the value of $\frac{F}{A}$ is :



(A) $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$

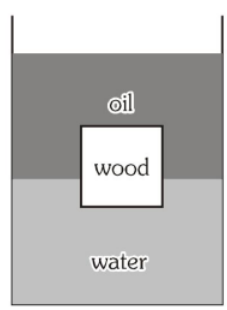
(B) $(\rho_1 h_1 + \rho_2 \ell_2 + \rho_3 h_3)g$

(C) $(\rho_2 h_2 + \rho_3 h_3)g$

(D) $(\rho_2 \ell_2 + \rho_3 h_3)g$

Buoyant Force

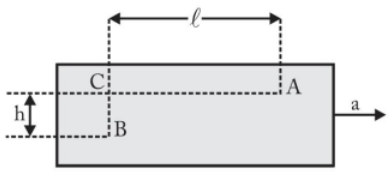
Q5. A cubical block of wood of side of 10 cm, floats at the interface between oil and water as shown in figure with its lower face 2 cm below the interface. The density of oil is 0.6 g/cm^3. The mass of the block is -



(A) 600 g (B) 680 g

(C) 800 g (D) 200 g

Q6. A sealed tank containing a liquid of density ρ moves with a horizontal acceleration a, as shown in the figure. The difference in pressure between the points A and B is -



(A) hρg (B) ℓρa
(C) hρg – ℓρa (D) hρg + ℓρa

Archimedes Principle

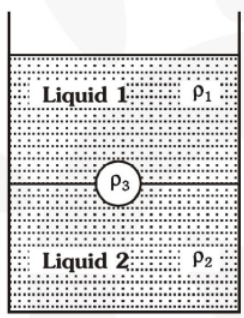
Q7. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2, respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 & ρ_3.

(A) $\rho_3 < \rho_1 < \rho_2$

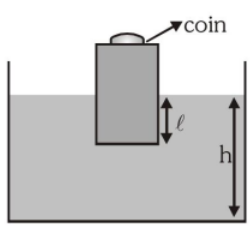
(B) $\rho_1 > \rho_3 > \rho_2$

(C) $\rho_1 < \rho_2 < \rho_3$

(D) $\rho_1 < \rho_3 < \rho_2$



Q8. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distance ℓ and h are shown there. After sometime the coin falls into the water. Then:-



(A) ℓ decreases and h increases
(B) ℓ increases and h decreases
(C) both ℓ and h increase
(D) both ℓ and h decrease

Q9. A piece of ice is floating in a jar containing water. When the ice melts, then the level of water :-

(A) Rises
(B) Falls
(C) Remains unchanged
(D) Changes erratically

Q10. A U-tube is partially filled with water. Oil which does not mix with water is next poured into one side, until water rises by 25 cm on the other side. If the density of the oil is 0.8 g/cc, the oil level will stand higher than the water level by :-

(A) 6.25 cm (B) 12.50 cm
(C) 18.75 cm (D) 25.00 cm

Q11. Two solids A and B of equal shape and size float in water. It is observed that A floats with half its volume immersed and B floats with 2/3 of its volume immersed. Compare the densities of A and B-

(A) 4 : 3 (B) 2 : 3
(C) 3 : 4 (D) 1 : 3

Eureka-I Found it

Q12. **Statement–1 :** A solid sphere and a hollow sphere of same material are floating in a liquid. Radius of both the spheres are same. Percentage of volume immersed of both the spheres will be same.

Statement–2 : Upthrust acts on volume of liquid displaced. It has nothing to do whether the body is solid or hollow.

(A) Statement–1 is true, Statement–2 is true; Statement–2 is correct explanation for Statement–1.

(B) Statement–1 is true, Statement–2 is true; Statement–2 is NOT a correct explanation for Statement–1.

(C) Statement–1 is true, Statement–2 is false.

(D) Statement–1 is false, Statement–2 is true.

Fluid Dynamics

Q13. Two water pipes P and Q having diameter 2×10^{-2} m and 4×10^{-2} m respectively are joined in series with the main supply line of water. The velocity of water flowing in pipe P is

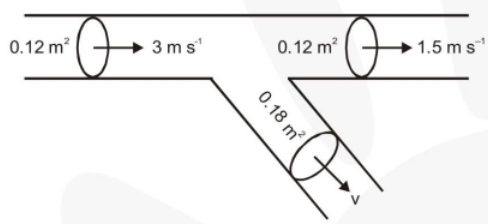
(A) 4 times that of Q

(B) 2 times that of Q

(C) $\frac{1}{2}$ times that of Q

(D) $\frac{1}{4}$ times that of Q

Q14. An incompressible liquid travels as shown in figure. Calculate the speed of the fluid in lower branch.

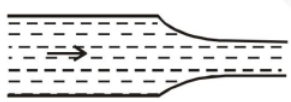


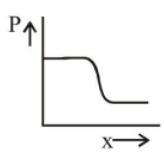
(A) 1 ms^{-1} (B) 2 ms^{-1}

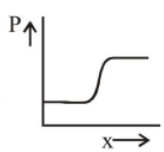
(C) 3 ms^{-1} (D) 4 ms^{-1}

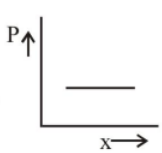
Bernoulli's Principle

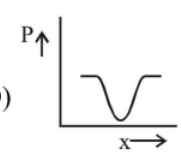
Q15. Water flows through a frictionless duct with a cross-section varying as shown in figure. Pressure P at points along the axis is represented by



(A) 

(B) 

(C) 

(D) 

Application of Bernoulli's Principle

Q16. The cylindrical tube of a spray pump has a radius R, one end of which has n fine holes, each of radius r. If the speed of flow of the liquid in the tube is v, the speed of ejection of the liquid through the hole is :

(A) $\frac{v}{n}\left(\frac{R}{r}\right)$ (B) $\frac{v}{n}\left(\frac{R}{r}\right)^{1/2}$

(C) $\frac{v}{n}\left(\frac{R}{r}\right)^{3/2}$ (D) $\frac{v}{n}\left(\frac{R}{r}\right)^{2}$

Speed of Efflux

Q17. A tank full of water has a small hole at its bottom. Let t_1 be the time taken to empty first one third of the tank and t_2 be the time taken to empty second one third of the tank and t_3 be the time taken to empty rest of the tank then :-

(A) $t_1 = t_2 = t_3$ (B) $t_1 > t_2 > t_3$

(C) $t_1 < t_2 < t_3$ (D) $t_1 > t_2 < t_3$

Viscosity

Q18. Poise is the unit of :-

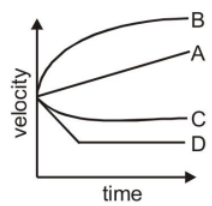
(A) pressure

(B) friction

(C) surface tension

(D) co-efficient of viscosity

Q19. A small ball is left in a viscous liquid from very much height. Correct graph of its velocity with time is :



(A) A (B) B

(C) C (D) D

Stokes' Law

Q20. A small drop of water falls from rest through a large height h in air experiencing viscous force. The final velocity is -

(A) almost independent of h

(B) proportional to $\sqrt{h}$

(C) proportional to h

(D) inversely proportional to h

Surface Tension

Q21. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of 1.5×10^{-2} N (see figure). The length of the slider is 30 cm and its weight negligible. The surface tension of the liquid film is :-



(A) 0.025 Nm^{-1} (B) 0.0125 Nm^{-1}

(C) 0.1 Nm^{-1} (D) 0.05 Nm^{-1}

Q22. A ring of radius r and weight W is lying on a liquid surface. If the surface tension of the liquid is T, then the minimum force required to be applied in order to lift the ring up -

(A) W (B) 2W

(C) $W + 4\pi rT$ (D) $W + 2\pi rT$

Excess Pressure in Drops and Bubbles

Q23. A water drop is divided into 8 equal droplets. The pressure difference between inner and outer sides of the big drop :-

(A) will be the same as for smaller droplet

(B) will be half of that for smaller droplet

(C) will be one-forth of that for smaller droplet

(D) will be twice of that for smaller droplet

Contact Angle

Q24. Shape of meniscus for a liquid of zero angle of contact is :-

(A) plane

(B) parabolic

(C) hemi-spherical

(D) cylindrical

Rise of Liquid in capillary

Q25. In a surface tension experiment with a capillary tube water rises up to 0.1 m. If the same experiment is repeated on an artificial satellite which is revolving round the earth, water will rise in the capillary tube up to a height of :-

(A) 0.1 m

(B) 0.98 m

(C) 9.8 m

(D) full length of capillary tube

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A beaker is filled in with water is accelerated a m/s^2 in + x direction. The surface of water shall make an angle :

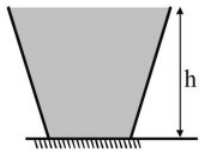
(A) $\tan^{-1}(a/g)$ backwards

(B) $\tan^{-1}(a/g)$ forwards

(C) $\cot^{-1}(g/a)$ backwards

(D) $\cot^{-1}(g/a)$ forwards

Q2. M gm of a liquid of density ρ is filled in a light beaker and kept on a horizontal table as shown in the figure. The height of the liquid in the beaker is h. The beaker is wider on top than at its base and the cross-sectional area of the base is A. Neglect the effect of atmospheric pressure. Now, choose the **CORRECT** statement(s) from the following.



(A) The pressure of liquid at the bottom surface is ρgh.

(B) The normal reaction exerted by the table on the beaker is ρghA.

(C) The pressure of the liquid at the bottom surface is $\frac{Mg}{A}$.

(D) The normal reaction exerted by the table on the beaker is Mg.

Q3. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2 :-

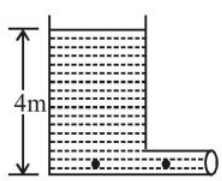
(A) the fraction of volume immersed in the liquid will change.

(B) the buoyancy force on the block will change.

(C) the buoyancy force will increase by 50 percent.

(D) the pressure in the liquid will increased.

Q4. A vent tank of large cross-sectional area has a horizontal pipe 0.12 m in diameter at the bottom. This holds a liquid whose density is 1500 kg/m^3 to a height of 4.0 m. Assume the liquid is an ideal fluid in laminar flow. In figure, the velocity with which fluid flows out is **:-**



(A) $2\sqrt{5}$ m/s

(B) $\sqrt{5}$ m/s

(C) $4\sqrt{5}$ m/s

(D) $\sqrt{10}$ m/s

Q5. A Newtonian fluid fills the clearance between a shaft and a sleeve. When a force of 800N is applied to the shaft, parallel to the sleeve, the shaft attains a speed of 1.5 cm/sec. If a force of 2.4 kN is applied instead, the shaft would move with a speed of :-

(A) 1.5 cm/sec (B) 13.5 cm/sec

(C) 4.5 cm/sec (D) None

Q6. If two soap bubbles of different radii are connected by a tube,

(A) air flows from the bigger bubble to the smaller bubble till the sizes become equal

(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged

(C) air flows from the smaller bubble to the bigger

(D) there is no flow of air.

Q7. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble:

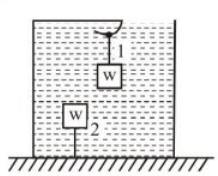
(A) is greater than the pressure outside it

(B) is less than the pressure outside it

(C) increases as the bubble moves up

(D) remains same as the bubble moves up

Q8. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.



(A) iron block gets deflected towards right.

(B) wooden block gets deflected towards right.

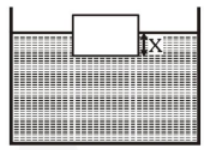
(C) iron block gets deflected towards left.

(D) wooden block gets deflected towards left.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. A cube is floating in a liquid as shown in figure.



Column-I	Column-II
(A) If density of liquid decreases then x will	(P) increase
(B) If size of cube is increased then x will	(Q) decrease
(C) If the whole system is accelerated upwards then x will	(R) remains same

Q10. Two soap bubbles coalesce to form a single large bubble :

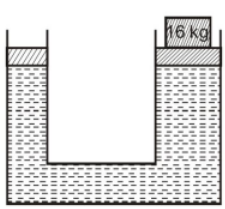
Column-I	Column-II
(A) Surface energy in the process will	(P) increase
(B) Temperature of the bubble will	(Q) decrease
(C) Pressure inside the soap bubble will	(R) remains same

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. A mercury barometer reads 75 cm in a stationary lift. What reading does it show when the lift is moving downwards, with an acceleration of 1 m/s^2 ?

Q12. The area of cross-section of wider tube is 800 cm^2. If a mass of 16 kg is placed on the massless piston, find the difference in the level of water in the two tubes.



SUBJECTIVE TYPE

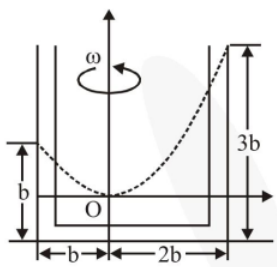
Q13. A certain block weighs 15N in air, it weighs 12N when immersed in water. When immersed in another liquid, it weighs 13N, calculate the relative density of (i) the block (ii) the other liquid.

Q14. A large wooden plate of area $10m^2$ floating on the surface of a river is made to move horizontally with a speed of 2 m/s by applying a tangential force. If the river is 1 m deep and the water in contact with the bed is stationary, find the tangential force needed to keep the plate moving. Coefficient of viscosity of water at the temperature of the river = 10^{-2} poise.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

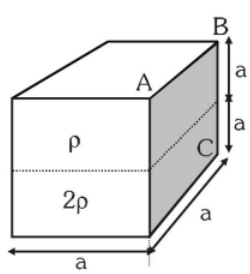
Q1. A vertical U-shaped tube having a liquid rotates about the vertical axis. The value of ω is :



(A) $\sqrt{\frac{g}{b}}$ (B) $2\sqrt{\frac{g}{3b}}$

(C) $\sqrt{\frac{g}{2b}}$ (D) none of these

Q2. A cuboid (a × a × 2a) is filled with two immiscible liquids of density 2ρ & ρ as shown in the figure. Neglecting atmospheric pressure, ratio of force on base & side wall of the cuboid is :-



(A) 2 : 3 (B) 1 : 3

(C) 5 : 6 (D) 6 : 5

Q3. What is the pressure 200 m below the surface of the ocean if the sp. gravity of sea water is 1.03

[Atmospheric pressure = 1.013×10^5 N/m^2].

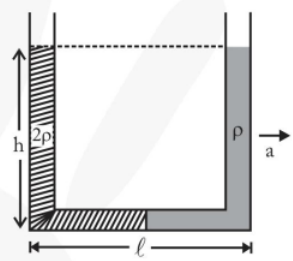
(A) 21.2×10^5 N/m^2

(B) 20.4×10^5 N/m^2

(C) 40.0×10^4 N/m^2

(D) 21.2×10^6 N/m^2

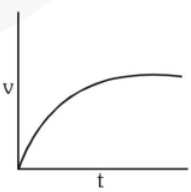
Q4. A U–tube of base length 'ℓ' filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration 'a' on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height h is given by -

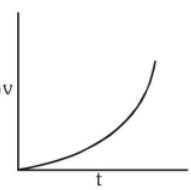


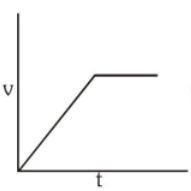
(A) $\frac{a}{2g}\ell$ (B) $\frac{3a}{2g}\ell$

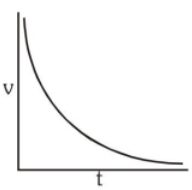
(C) $\frac{a}{g}\ell$ (D) $\frac{2a}{3g}\ell$

Q5. A piece of cork starts from rest at the bottom of a lake and floats up. Its velocity v is plotted against time t. Which of the following best represents the resulting curve-

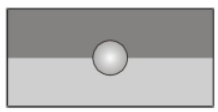
(A) 

(B) 

(C) 

(D) 

Q6. A solid uniform ball having volume V and density ρ floats at the interface of two immiscible liquids as shown in figure. The densities of the upper and the lower liquids are ρ_1 and ρ_2 respectively, such that $\rho_1 < \rho < \rho_2$. What fraction of the volume of the ball will be in the lower liquid :-



(A) $\frac{\rho-\rho_2}{\rho_1-\rho_2}$ (B) $\frac{\rho_1}{\rho_1-\rho_2}$

(C) $\frac{\rho_1-\rho}{\rho_1-\rho_2}$ (D) $\frac{\rho_1-\rho_2}{\rho_2}$

Q7. A wooden ball of density D is immersed in water of density d to a depth h below the surface of water. Upto which height the ball will jump out of water, when it is released ?

(A) $\frac{d}{D}h$ (B) $\left(\frac{d}{D}-1\right)h$

(C) h (D) Zero

Q8. A steel ball is floating in a trough of mercury. If we fill the empty part of the trough with water, what will happen to the steel ball :-

(A) It will continue in its position

(B) It will move up

(C) It will move down

(D) It will execute vertical oscillations

Q9. A wooden block floats in water in a sealed container. When the container is at rest, 25% of the block is above the water. Consider the following five situation :

(i) the container is lifted up at constant speed.

(ii) the container is lowered at constant speed.

(iii) the container is lifted up at an increasing speed.

(iv) the container is lowered at a decreasing speed.

(v) the air pressure above the water in the container is increased.

What happens in each situation?

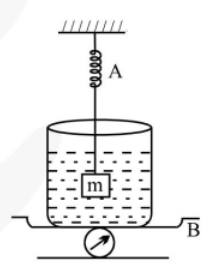
(A) 75% of block is submerged in siutation (i) & (ii) and higher fraction is submerged in situations (iii), (iv) & (v).

(B) 75% of block is submerged in situation (i), (ii) & (v) and higher fraction is submerged in situations (iii), (iv).

(C) 75% of block is submerged in situation (i), (ii), (iii) & (iv) and higher fraction is submerged in situations (v).

(D) 75% of block is submerged in all situation.

Q10. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation :



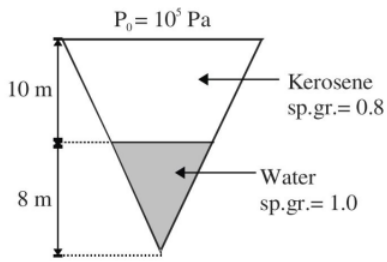
(A) the balance A will read more than 2 kg

(B) the balance B will read more than 5 kg

(C) the balance A will read less than 2 kg and B will read less than 5 kg

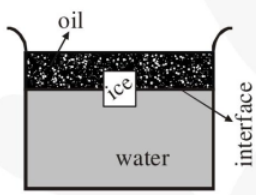
(D) the balances A and B will read 2 kg and 5 kg respectively.

Q11. Shown in figure, a conical container of half-apex angle 37° filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is approximately, (Take atmospheric pressure = 10^5 Pa).



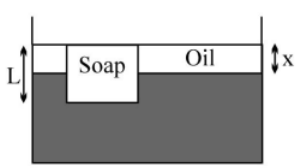
(A) 3×10^7 N (B) 4×10^7 N
(C) 2×10^7 N (D) 5×10^7 N

Q12. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively:-



(A) rise and fall
(B) fall and rise
(C) rise and not change
(D) not change and fall

Q13. A rectangular bar of soap has density 800 kg/m³ floats in water of density 1000 kg/m³. Oil of density 300 kg/m³ is slowly added, forming a layer that does not mix with the water. When the top surface of the oil is at the same level as the top surface of the soap. What is the ratio of the oil layer thickness to the soap's thickness, x/L ?



(A) $\frac{2}{10}$ (B) $\frac{2}{7}$ (C) $\frac{3}{10}$ (D) $\frac{3}{8}$

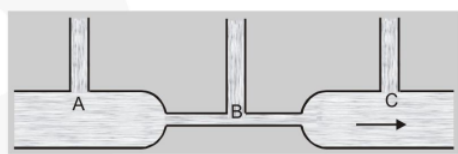
Q14. The total weight of a piece of wood is 6 kg. In the floating state in water its $\frac{1}{3}$ part remains inside the water. On this floating solid, what maximum weight is to be put such that the whole of the piece of wood is to be drowned in the water :-

(A) 12 kg (B) 10 kg
(C) 14 kg (D) 15 kg

Q15. Water is flowing steadily through a horizontal tube of non uniform cross-section. If the pressure of water is 4×10^4 N/m² at a point where cross-section is 0.02 m² and velocity of flow is 2 m/s, what is pressure at a point where cross-section reduces to 0.01 m².

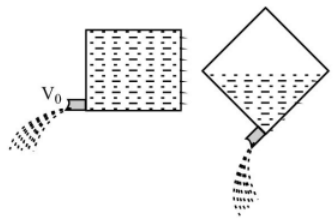
(A) 1.4×10^4 N/m² (B) 3.4×10^4 N/m²
(C) 2.4×10^{-4} N/m² (D) none of these

Q16. In the figure below is shown the flow of liquid through a horizontal pipe. Three tubes A, B and C are connected to the pipe. The radii of the tubes, A, B and C at the junction are respectively 2 cm, 1 cm. and 2 cm. It can be said that the :-



(A) height of the liquid in the tube A is maximum
(B) height of the liquid in the tubes A and B is the same
(C) height of the liquid in all the three tubes is the same
(D) height of the liquid in the tubes A and C is the same

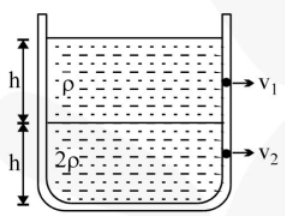
Q17. A cubical box of wine has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of wine with a initial speed of v_0 (see figure). When the box is half empty, someone tilts it at 45° so that the spout is at the lowest point (see figure). When the spout is opened the wine will flow out with a speed of :



(A) v_0 (B) $v_0/2$

(C) $v_0/\sqrt{2}$ (D) $v_0/\sqrt[4]{2}$

Q18. Equal volumes of two immiscible liquids of densities ρ and 2ρ are filled in a vessel as shown in figure. Two small holes are punched at depth h/2 and 3h/2 from the surface of lighter liquid. If v_1 and v_2 are the velocities of a efflux at these two holes, then v_1/v_2 is :



(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{2}}$

Q19. Water from a tap emerges vertically downwards with an initial speed of 1.0 m/s. The cross–sectional area of tap is 10^{-4} m^2. Assume that the pressure is constant throughout the stream of water and that the flow is steady, the cross–sectional area of stream 0.15 m below the tap is :–

(A) 5.0×10^{-4} m^2

(B) $1.0 \times \times 10^{-4}$ m^2

(C) $5.0 \times \times 10^{-5}$ m^2

(D) $2.0 \times \times 10^{-5}$ m^2

Q20. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth 4y from the top. When the tank is completely filled with water, the quantities of water flowing out per second from the holes are both same. Then, R is equal to :-

(A) $\frac{L}{\sqrt{2\pi}}$ (B) $2\pi L$ (C) L (D) $\frac{L}{2\pi}$

Q21. A cylindrical vessel is filled with water up to height H.A hole is bored in the wall at a depth h from the free surface of water. For maximum range, h is equal to :-

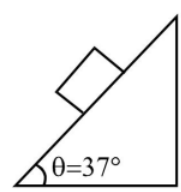
(A) $\frac{H}{4}$ (B) $\frac{H}{2}$ (C) $\frac{3H}{4}$ (D) H

Q22. A cylindrical vessel of cross-sectional area 1000 cm^2 is fitted with a frictionless piston of mass 10 kg and filled with water completely. A small hole of cross-sectional area 10 mm^2 is opened at a point 50 cm deep from the lower surface of the piston. The velocity of efflux from the hole will be :-

(A) 10.5 m/s (B) 3.4 m/s

(C) 0.8 m/s (D) 0.2 m/s

Q23. A cubical block of side 'a' and density 'ρ' slides over a fixed inclined plane with constant velocity 'v'. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be:



(A) $\frac{3\rho a g t}{5v}$ (B) $\frac{4\rho a g t}{5v}$

(C) $\frac{\rho a g t}{v}$ (D) none of these

Q24. If a ball of steel (density $\rho = 7.8$ g cm^{-3}) attains a terminal velocity of 10 cm s^{-1} when falling in a tank of water (coefficient of viscosity $\eta_{water} = 8.5 \times 10^{-4}$ Pa–s) then its terminal velocity in glycerine ($\sigma = 1.2$ g cm^{-3}, $\eta_{gly} = 13.2$ Pa–s) would be nearly :-

(A) 1.6×10^{-5} cm s^{-1}

(B) 3.98×10^{-4} cm s^{-1}

(C) 6.25×10^{-4} cm s^{-1}

(D) 1.5×10^{-5} cm s^{-1}

Q25. Two merucry drops (each of radius 'r') merge to form a bigger drop. The surface energy of the bigger drop, if $\frac{1}{\pi}$ is the surface tension (in SI unit), is :

(A) $2^{5/3} r^2$ (B) $4r^2$

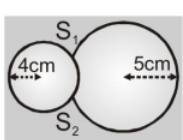
(C) $2r^2$ (D) $2^{8/3} r^2$

Q26. Two drops of same radius are falling through air with steady velocity of v cm/s. If the two drops collapse, what would be the terminal velocity ?

(A) 4 v (B) $(4)^{1/3}$ v

(C) 2 v (D) 64 v

Q27. Two soap bubbles of radii r_1 and r_2 equal to 4cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be :-



(A) 4 cm (B) 20 cm

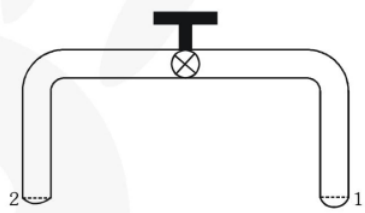
(C) 5 cm (D) 4.5 cm

Q28. If a capillary of radius r is dipped in water, the height of water that rises in it is h and its mass is M. If the radius of the capillary is doubled the mass of water that rises in the capillary will be

(A) 4M (B) 2M

(C) M (D) $\frac{M}{2}$

Q29. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r. End 2 has sub–hemispherical soap bubble as shown in figure. Just after opening the valve,



(A) Air from end 1 flows towards end 2. No change in the volume of the soap bubbles.

(B) Air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases.

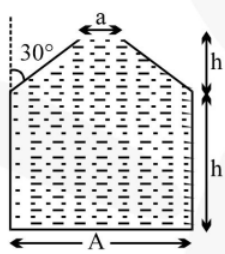
(C) No change occurs

(D) Air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases.

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h. The upper part is a conical vessel of height h with base area 'A' and top area 'a' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height 2h. Neglecting atmospheric pressure.



(A) The force F exerted by the liquid on the base of the vessel is $2h\rho g\,\frac{(A+a)}{2}$

(B) the pressure P at the base of the vessel is $2h\rho g\,\frac{A}{a}$

(C) the weight of the liquid W is greater than the force exerted by the liquid on the base

(D) the walls of the vessel exert a downward force on the liquid.

Q2. A heavy hollow cone of radius R and height h is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density ρ. The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure, find the total upward force exerted by water on the cone is -

(A) $(2/3)\pi R^2h\rho g$ (B) $(1/3)\pi R^2h\rho g$

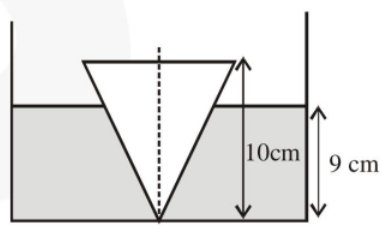
(C) $\pi R^2h\rho g$ (D) None

Q3. A solid floats with 2/3 of its volume immersed in a liquid and with 3/4 of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture formed of equal volumes of the liquids?

(A) 6/7 (B) 8/11

(C) 11/16 (D) 12/17

Q4. An inverted ice cone of height 10 cm is immersed in water as shown below. Tip of cone is touching the base of vessel filled with water upto a height of 9 cm. (Specific Gravity of ice is 0.9)



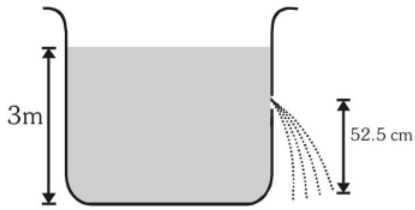
(A) Normal contact force between cone & vessel is zero.

(B) After melting of ice, the level of water will increase.

(C) Potential energy of ice–liquid system will decrease after melting of ice.

(D) Cone is in stable equilibrium.

Q5. Water is filled in a cylindrical container to a height of 3 m. The ratio of the cross–sectional area of the orifice and the beaker is 0.1. The square of the speed of the liquid coming out from the orifice is ($g = 10\ m/s^2$)

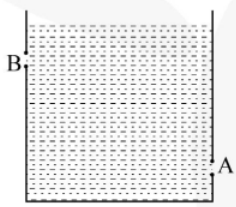


(A) 50 m^2s^2 (B) 50.5 m^2s^2
(C) 51 m^2s^2 (D) 52 m^2s^2

Q6. A small ball of relative density 0.8 falls into water from a height of 2m. The depth to which the ball will sink is (neglect viscous forces) :

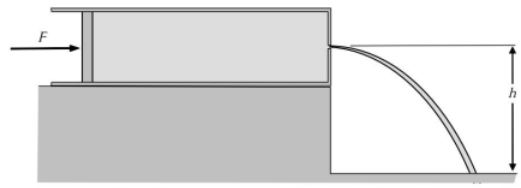
(A) 8 m (B) 2 m
(C) 6 m (D) 4 m

Q7. Two identical holes each of cross-sectional area 10^{-3} m^2 are made on the opposite sides of a tank containing water as shown in the figure. As the water comes out of the holes, the tank will experience a net horizontal force of 20 N. The difference in height between the holes A and B is -



(A) 1 m (B) 0.5 m
(C) 2 m (D) 0.25 m

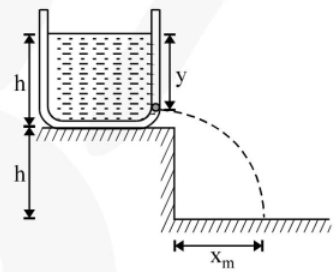
Q8. An ideal liquid of density ρ is filled in a horizontally fixed syringe fitted with piston. There is no friction between the piston and the inner surface of the syringe. Cross-section area of the syringe is A. At one end of the syringe, an orifice of negligible cross-section area is made. When the piston is pushed into the syringe, the liquid comes out of the orifice following parabolic path and falls on the ground. With what speed the liquid strikes the ground? Neglect the air drag :-



(A) $\sqrt{\frac{F+\rho ghA}{\rho A}}$ (B) $\sqrt{\frac{F+2\rho ghA}{\rho A}}$

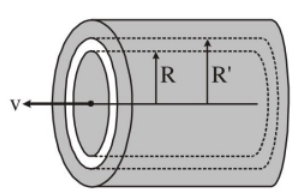
(C) $\sqrt{\frac{2F+\rho ghA}{\rho A}}$ (D) $\sqrt{\frac{2(F+\rho ghA)}{\rho A}}$

Q9. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then -



(A) $x_m = 2h$ (B) $x_m = 1.5\ h$
(C) $y = h$ (D) $y = 0.75\ h$

Q10. Figure shows a long solid cylinder of radius R inside a long hollow tube of inner radius R′. A viscous liquid of coefficient of viscosity η is filled in the gap between the cylinder and the hollow tube. If the solid cylinder is moved with velocity v parallel to its length inside the fixed hollow tube, find the force required per unit length to be applied on the solid cylinder, assuming uniform velocity gradient in the liquid.



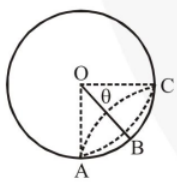
(A) $\eta(2\pi R')\left(\frac{V}{R'-R}\right)$

(B) $\eta(2\pi R)\left(\frac{V}{R'-R}\right)$

(C) $\eta(4\pi R')\left(\frac{V}{R'+R}\right)$

(D) $\eta(4\pi R)\left(\frac{V}{R'+R}\right)$

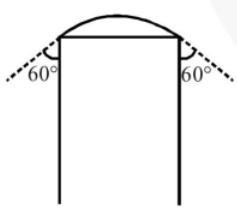
Q11. Consider a small water drop in air. If T is the surface tension, then what is the force due to surface tension acting on the smaller section ABC ?



(A) $2TR\pi$ (B) $2\pi TR\sin\theta$

(C) $2\pi TR\sin^2\theta$ (D) $2\pi TR\sin^3\theta$

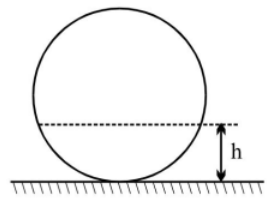
Q12. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of 60° with the tube as shown. The excess of pressure over the atmospheric pressure in the tube is :



(A) 5 Pa (B) 1 Pa

(C) 10 Pa (D) 20 Pa

Q13. A liquid is filled in a spherical container of radius R till a height h. At this position the liquid surface at the edges is also horizontal. The contact angle is :-



(A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$

(C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$

Q14. When a capillary tube is dipped in a liquid, the liquid rises to a height h in the tube. The free liquid surface inside the tube is hemispherical in shape. The tube is now pushed down so that the height of the tube outside the liquid is less than h :-

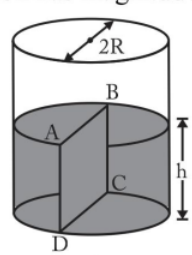
(A) The liquid will come out of the tube like in a small fountain.

(B) The liquid will ooze out of the tube slowly.

(C) The liquid will fill the tube but not come out of its upper end.

(D) The free liquid surface inside the tube will not be hemispherical.

Q15. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is r, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude :–



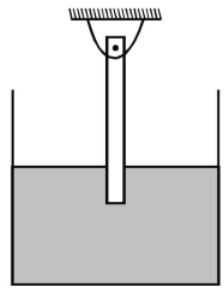
(A) $[2P_0Rh + \pi R^2\rho gh - 2RT]$

(B) $[2P_0Rh + R\rho gh^2 - 2RT]$

(C) $[P_0\pi R^2 + R\rho gh^2 - 2RT]$

(D) $[P_0\pi R^2 + R\rho gh^2 + 2RT]$

Q16. A thin vertical uniform wooden rod is pivoted at the top and immersed in water as shown. The container is slowly raised. At a certain moment, the equilibrium becomes unstable. If density of water is 9/5 times the density of wood, then ratio of total length of rod to the submerged length of rod, at that moment is



(A) 2 (B) 3 (C) 4 (D) 6

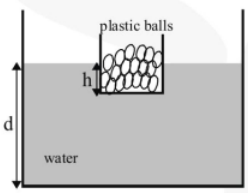
MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. Column I shows different system as describe, with some parameter while column-II gives the change in the parameter.

Column-I **Column-II**

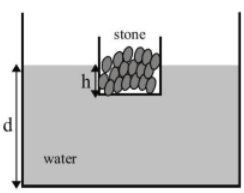
(A)



(P) h decreases

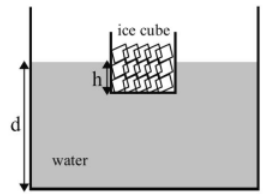
Plastic ball is thrown from the container in the water.
[d is the height of water level while h is the depth to which the container is submerged]

(B)



(Q) h remain same

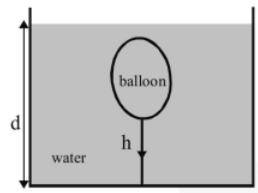
Stone is thrown from the container in the water.
[d is the height of water level while h is the depth to which the container is submerged]

(C) 

(R) h increases

Ice melts and remain in the container.

[d is the height of water level while h is the depth to which the container is submerged]

(D) 

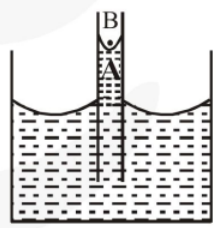
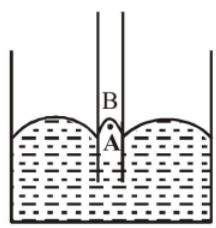
(S) d decreases

The water is heated slowly.

[d is the height of water level while h is the tension in the string. $\gamma_{air} > \gamma_{water}$]

(T) d remain same

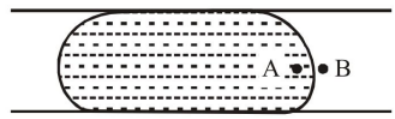
Q18. Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Point A and B are just inside and outside the surface. Match the following

Column-I	**Column-II**
(A) Liquid is wetting the solid	(P) 
(B) Liquid is not wetting the solid	(Q) 

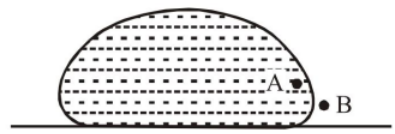
(C) Pressure at A > pressure at B

(D) Pressure at B > Pressure at A

(R) A liquid drop is pressed between two parallel glass plates.



(S)

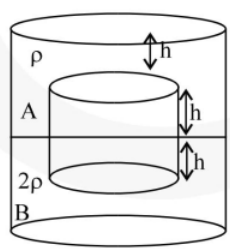


(T)



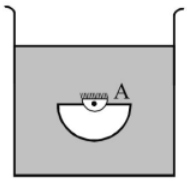
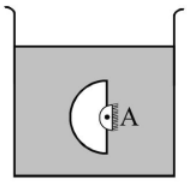
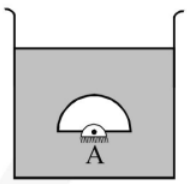
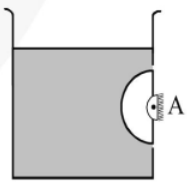
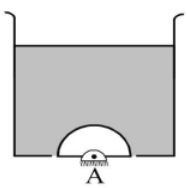
both end open cylindrical pipe

Q19. Shown below is a cylinder of radius R floating in vessel containing liquids A and B. Neglecting atmospheric pressure match the quantities mentioned in column-I with corresponding expression in column-II.



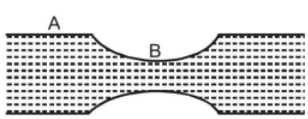
Column-I	**Column-II**
(A) Net force exerted by liquid A of density ρ on the cylinder.	(P) $9\ \rho g R h^2$
(B) Net force exerted by liquid B of density 2ρ on the cylinder.	(Q) $\pi\rho g R^2 h$
(C) Net force exerted by liquids A and B on the left half of the curved part of cylinder.	(R) $4\pi\rho g R^2 h$
(D) Net force exerted by liquid A and B on the cylinder.	(S) $3\pi\rho g R^2 h$

Q20. In column-II semi-cylinder of material having density less than fluid is hinged about point A

Column-I	**Column-II**
(A) Torque is clockwise.	(P) 
(B) Effective force due to fluid is at COM of body	(Q) 
(C) Body in stable equilibrium.	(R) 
(D) Force due to fluid produces no torque about axis.	(S) 
	(T) 

SUBJECTIVE TYPE

Q21. Water flows in a horizontal tube as shown in figure. The pressure of water changes by 600 N/m^2 between A and B where the area of cross-section are 30 cm^2 and 15 cm^2 respectively. Find the rate of flow of water through the tube.



Q22. A liquid drop of radius 'r' is divided into 64 similar tiny droplets. If the surface tension of liquid is T. Calculate the increase in surface energy ?

Q23. On dipping one end of a capillary in liquid and inclining the capillary at an angles 30° and 60° with the vertical, the lengths of liquid columns in it are found to be ℓ_1 and ℓ_2 respectively. Find the ratio of ℓ_1 and ℓ_2 ?

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	C	C	C	B	D	D	D	C	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	A	A	A	D	C	D	C	A
21	**22**	**23**	**24**	**25**					
A	C	B	C	D					

Ex-1A

1. A,C **2.** A,D **3.** B, C, D **4.** C **5.** C

6. C **7.** A **8.** B, C **9.** (A) P, (B) P, (C) R

10. (A) Q, (B) P, (C) Q **11.** 83.33 cm **12.** 20 cm **13.** 5, 0.67

14. 2×10^{-2} N

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**D**	**A**	**B**	**A**	**C**	**B**	**B**	**D**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**A**	**B**	**A**	**B**	**D**	**D**	**D**	**C**	**A**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	
B	**B**	**A**	**C**	**D**	**B**	**B**	**B**	**B**	

Ex-2A

1. D **2.** A **3.** D **4.** B,C **5.** A

6. A **7.** A **8.** D **9.** A,C **10.** B

11. C **12.** C **13.** B **14.** C,D **15.** B

16. B

17. (A)-P,T; (B)-P,S; (C)-Q,T; (D)-R

18. (A)-P,T; (B)-Q,R,S; (C)-Q,R,S,T; (D)-P

19. (A)-Q, (B)-R, (C)-P, (D)-S

20. (A) $\rightarrow$(Q); (B) $\rightarrow$(PQR); (C) $\rightarrow$(R); (D) $\rightarrow$(PRST)

21. $\frac{6}{\sqrt{10}} \times 10^{-3}\ m^3/s$ **22.** $12\pi r^2 T$ **23.** $\frac{1}{\sqrt{3}}$

-: Important Notes & Formulas :-